

HYPERTENSION

Key facts

- An estimated 1.4 billion adults aged 30–79 years worldwide had hypertension in 2024; this represents 33% of the population in this age range.
 - Two-thirds of adults aged 30–79 years who have hypertension live in low- and middle-income countries.
 - An estimated 600 million adults with hypertension (44%) are unaware that they have the condition.
 - Approximately 630 million adults with hypertension (44%) are diagnosed and treated.
 - Approximately 320 million adults with hypertension (23%) have it under control.
 - Hypertension is a major cause of premature death worldwide.
 - One of the global targets for noncommunicable diseases is to reduce the prevalence of uncontrolled hypertension by 25% between 2010 and 2025.
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Overview

Hypertension (high blood pressure) is when the pressure in your blood vessels is too high (140/90 mmHg or higher). It is common but can be serious if not treated.

People with high blood pressure may not feel symptoms. The only way to know is to get your blood pressure checked.

Things that increase the risk of having high blood pressure include:

- older age
- genetics
- being overweight or obese
- not being physically active
- high-salt diet
- drinking too much alcohol

Lifestyle changes like eating a healthier diet, quitting tobacco and being more active can help lower blood pressure. Some people may still need to take medicines.

Blood pressure is written as two numbers. The first (systolic) number represents the pressure in blood vessels when the heart contracts or beats. The second (diastolic)

number represents the pressure in the vessels when the heart rests between beats.

Hypertension is diagnosed if, when it is measured on two different days, the systolic blood pressure readings on both days is ≥ 140 mmHg and/or the diastolic blood pressure readings on both days is ≥ 90 mmHg.

Risk factors

Modifiable risk factors include unhealthy diets (excessive salt consumption, a diet high in saturated fat and trans fats, low intake of fruits and vegetables), physical inactivity, consumption of tobacco and alcohol, and being overweight or obese. In addition, there are environmental risk factors for hypertension and associated diseases, where air pollution is the most significant.

Non-modifiable risk factors include a family history of hypertension, age over 65 years and co-existing diseases such as diabetes or kidney disease.

Symptoms

Most people with hypertension don't feel any symptoms. Very high blood pressures can cause headaches, blurred vision, chest pain and other symptoms.

Checking your blood pressure is the best way to know if you have high blood pressure. If hypertension isn't treated, it can cause other health conditions like kidney disease, heart disease and stroke.

People with very high blood pressure (usually 180/120 or higher) can experience symptoms including:

- severe headaches
- chest pain
- dizziness
- difficulty breathing
- nausea
- vomiting
- blurred vision or other vision changes
- anxiety
- confusion
- buzzing in the ears
- nosebleeds

- abnormal heart rhythm

If you are experiencing any of these symptoms and a high blood pressure, seek care immediately.

The only way to detect hypertension is to have a health professional measure blood pressure. Having blood pressure measured is quick and painless. Although individuals can measure their own blood pressure using automated devices, an evaluation by a health professional is important for assessment of risk and associated conditions.

Treatment

Lifestyle changes can help lower high blood pressure. These include:

- eating a healthy, low-salt diet
- losing weight
- being physically active
- quitting tobacco.

If you have high blood pressure, your doctor may recommend one or more medicines. Your recommended blood pressure goal may depend on what other health conditions you have.

Blood pressure goal is less than 130/80 if you have:

- cardiovascular disease (heart disease or stroke)
- diabetes (high blood sugar)
- chronic kidney disease
- high risk for cardiovascular disease.

For most people, the goal is to have a blood pressure less than 140/90.

There are several common blood pressure medicines:

- ACE inhibitors including enalapril and lisinopril relax blood vessels and prevent kidney damage.
- Angiotensin-2 receptor blockers (ARBs) including losartan and telmisartan relax blood vessels and prevent kidney damage.
- Calcium channel blockers including amlodipine and felodipine relax blood vessels.
- Diuretics including hydrochlorothiazide and chlorthalidone eliminate extra water from the body, lowering blood pressure.

Prevention

Lifestyle changes can help lower high blood pressure and can help anyone with hypertension. Many who make these changes will still need to take medicine.

These lifestyle changes can help prevent and lower high blood pressure.

Do:

- Eat more vegetables and fruits.
- Sit less.
- Be more physically active, which can include walking, running, swimming, dancing or activities that build strength, like lifting weights.
 - Get at least 150 minutes per week of moderate-intensity aerobic activity or 75 minutes per week of vigorous aerobic activity.
 - Do strength building exercises 2 or more days each week.
- Lose weight if you're overweight or obese.
- Take medicines as prescribed by your health care professional.
- Keep appointments with your health care professional.

Don't:

- eat too much salty food (try to stay under 2 grams per day)
- eat foods high in saturated or trans fats
- smoke or use tobacco
- drink too much alcohol (1 drink daily max for women, 2 for men)
- miss or share medication.

Reducing hypertension prevents heart attack, stroke and kidney damage, as well as other health problems.

Reduce the risks of hypertension by:

- reducing and managing stress
- regularly checking blood pressure
- treating high blood pressure
- managing other medical conditions

- reducing exposure to polluted air.

Complications of uncontrolled hypertension

Among other complications, hypertension can cause serious damage to the heart. Excessive pressure can harden arteries, decreasing the flow of blood and oxygen to the heart. This elevated pressure and reduced blood flow can cause:

- chest pain, also called angina;
- heart attack, which occurs when the blood supply to the heart is blocked and heart muscle cells die from lack of oxygen. The longer the blood flow is blocked, the greater the damage to the heart;
- heart failure, which occurs when the heart cannot pump enough blood and oxygen to other vital body organs; and
- irregular heart beat which can lead to a sudden death.

Hypertension can also burst or block arteries that supply blood and oxygen to the brain, causing a stroke.

In addition, hypertension can cause kidney damage, leading to kidney failure.

Prevalence of hypertension

The prevalence of hypertension varies across regions and country income groups. The WHO Eastern Mediterranean Region has the highest prevalence of hypertension (38%) while the WHO Western Pacific Region has the lowest prevalence of hypertension (29%).

The number of adults with hypertension increased from 650 million in 1990 to 1.4 billion in 2024, with the increase seen largely in low- and middle-income countries. This increase is due mainly to a rise in the number of older adults in those countries.

DIABETES

- The number of people living with diabetes rose from 200 million in 1990 to 830 million in 2022. Prevalence has been rising more rapidly in low- and middle-income countries than in high-income countries.

- More than half of people living with diabetes did not take medication for their diabetes in 2022. Diabetes treatment coverage was lowest in low- and middle-income countries.
- Diabetes causes blindness, kidney failure, heart attacks, stroke and lower limb amputation.
- In 2021, diabetes and kidney disease due to diabetes caused over 2 million deaths. In addition, around 11% of cardiovascular deaths were caused by high blood glucose.
- A healthy diet, regular physical activity, maintaining a normal body weight and avoiding tobacco use are ways to prevent or delay the onset of type 2 diabetes.
- Diabetes can be treated and its consequences avoided or delayed with diet, physical activity, medication and regular screening and treatment for complications.

Symptoms

Symptoms of diabetes may occur suddenly. In type 2 diabetes, the symptoms can be mild and may take many years to be noticed.

Symptoms of diabetes include:

- feeling very thirsty
- needing to urinate more often than usual
- blurred vision
- feeling tired
- losing weight unintentionally

Over time, diabetes can damage blood vessels in the heart, eyes, kidneys and nerves.

People with diabetes have a higher risk of health problems including heart attack, stroke and kidney failure.

Diabetes can cause permanent vision loss by damaging blood vessels in the eyes.

Many people with diabetes develop problems with their feet from nerve damage and poor blood flow. This can cause foot ulcers and may lead to amputation.

Type 1 diabetes

Type 1 diabetes (previously known as insulin-dependent, juvenile or childhood-onset) is characterized by deficient insulin production and requires daily administration of

insulin. In 2017 there were 9 million people with type 1 diabetes; the majority of them live in high-income countries. Neither its cause nor the means to prevent it are known.

Type 2 diabetes

Type 2 diabetes affects how your body uses sugar (glucose) for energy. It stops the body from using insulin properly, which can lead to high levels of blood sugar if not treated.

Over time, type 2 diabetes can cause serious damage to the body, especially nerves and blood vessels.

Type 2 diabetes is often preventable. Factors that contribute to developing type 2 diabetes include being overweight, not getting enough exercise, and genetics.

Early diagnosis is important to prevent the worst effects of type 2 diabetes. The best way to detect diabetes early is to get regular check-ups and blood tests with a healthcare provider.

Symptoms of type 2 diabetes can be mild. They may take several years to be noticed. Symptoms may be similar to those of type 1 diabetes but are often less marked. As a result, the disease may be diagnosed several years after onset, after complications have already arisen.

More than 95% of people with diabetes have type 2 diabetes. Type 2 diabetes was formerly called non-insulin dependent, or adult onset. Until recently, this type of diabetes was seen only in adults but it is now also occurring increasingly frequently in children.

Gestational diabetes

Gestational diabetes is hyperglycaemia with blood glucose values above normal but below those diagnostic of diabetes. Gestational diabetes occurs during pregnancy.

Women with gestational diabetes are at an increased risk of complications during pregnancy and at delivery. These women and possibly their children are also at increased risk of type 2 diabetes in the future.

Gestational diabetes is diagnosed through prenatal screening, rather than through reported symptoms.

Impaired glucose tolerance and impaired fasting glycaemia

Impaired glucose tolerance (IGT) and impaired fasting glycaemia (IFG) are intermediate conditions in the transition between normality and diabetes. People with IGT or IFG are at high risk of progressing to type 2 diabetes, although this is not inevitable.

Prevention

Lifestyle changes are the best way to prevent or delay the onset of type 2 diabetes.

To help prevent type 2 diabetes and its complications, people should:

- reach and keep a health body weight
- stay physically active with at least 150 minutes of moderate exercise each week
- eat a healthy diet and avoid sugar and saturated fat
- not smoke tobacco.

Diagnosis and treatment

Early diagnosis can be accomplished through relatively inexpensive testing of blood glucose. People with type 1 diabetes need insulin injections for survival.

One of the most important ways to treat diabetes is to keep a healthy lifestyle.

Some people with type 2 diabetes will need to take medicines to help manage their blood sugar levels. These can include insulin injections or other medicines. Some examples include:

- metformin
- sulfonylureas
- sodium-glucose co-transporters type 2 (SGLT-2) inhibitors.

Along with medicines to lower blood sugar, people with diabetes often need medications to lower their blood pressure and statins to reduce the risk of complications.

Additional medical care may be needed to treat the effects of diabetes:

- foot care to treat ulcers
- screening and treatment for kidney disease
- eye exams to screen for retinopathy (which causes blindness).

HEART AND KIDNEY DISEASE

If I have kidney disease, why should I learn about heart disease?

If you have kidney disease, you are more likely to get heart disease. Heart disease is the most common cause of death among people who have kidney disease. However, learning about kidney disease and heart disease can help you find ways to stay healthy.



If you have kidney disease, you are more likely to get heart disease.

What is kidney disease?

Kidney disease means that the kidneys are damaged and can't filter blood as they should. This damage can cause wastes to build up in the body. For most people, kidney damage occurs slowly over many years. This gradual loss of kidney function is called chronic kidney disease (CKD).

Most patients with CKD have no symptoms until kidney damage is advanced. After many years, you may start to feel sick or tired all of the time. Kidney failure is when you need a kidney transplant or a blood filtering treatment called dialysis to stay alive. Before you reach that stage, other health problems may develop. One of these problems is heart disease.

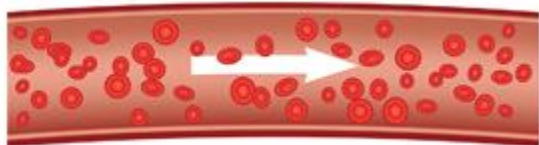
What is heart disease?

Heart disease includes any problem that keeps your heart from pumping blood as well as it should. The problem might start in your blood vessels or your heart. Heart and blood vessel problems include

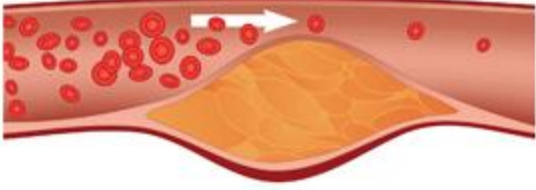
- the buildup of a substance called plaque in the walls of the blood vessels
- a blood clot that blocks the flow of blood to the heart
- heart attack—heart damage caused by a lack of blood and oxygen to the heart

The buildup of plaque is often the first step in making other problems. Plaque can block blood flow.

Plaque can block blood flow.



Normal blood vessel

	Blood vessel with plaque
	Blood vessel with plaque and blood clot

The early symptoms of plaque in your blood vessels include

- pain in your chest, called angina
- pain in your legs when walking
- sudden numbness or weakness in your arms or legs
- temporary signs of a stroke—a blockage of blood to the brain—such as having a hard time speaking or drooping muscles in your face
- feeling dizzy at times

A blood clot may form in a blood vessel that carries blood to the heart muscle. Then your heart muscle does not get the oxygen and nutrients it needs from the blood. The muscle becomes damaged. The damage to your heart caused by this blockage and lack of oxygen is called a heart attack.

Each person may have different heart attack symptoms. Symptoms can include

- chest pains or discomfort, or often a sense of chest pressure
- pain or discomfort in one or both arms—often the left arm—or in the back, jaw, neck, or stomach
- shortness of breath
- heavy sweating
- nausea or vomiting
- light-headedness

Women may not have chest pain. However, women may be more likely to have shortness of breath, nausea, or back and jaw pain.

If you have any of these symptoms, call 911 right away. You can start getting treatment in the ambulance on your way to the emergency room.

How are kidney disease and heart disease related?

Kidney disease and heart disease share two of the same main causes:

- [diabetes](#)
- [high blood pressure](#) *NIH external link*

Diabetes

If you have diabetes, you have too much glucose, also called sugar, in your blood. Too much glucose in your blood for a long time can damage many parts of your body, including your heart and kidneys. More information is provided in the following NIDDK health topics:

- [Diabetic Kidney Disease](#)
- [Diabetes, Heart Disease, and Stroke](#)

High Blood Pressure

Blood pressure is the force of your blood pushing against the walls of your blood vessels. With high blood pressure, your heart works harder to pump blood, which can strain your heart. High blood pressure can damage your blood vessels. If high blood pressure damages the small blood vessels in your kidneys, your kidneys will not filter your blood as well as they should.

Blood pressure is written as two numbers separated by a slash. A doctor will say a blood pressure reading of 120/80 as “120 over 80.” The top number represents the force of blood pressing against the blood vessel walls when the heart is contracting. The bottom number represents the pressure when the heart is at rest between contractions. You should keep your blood pressure below 140/90 unless your doctor sets a different goal. Controlling your blood pressure can help protect your heart and kidneys.

High blood pressure is not only a cause of kidney disease; kidney disease is also a cause of high blood pressure. When you have damaged kidneys, they may be unable to filter extra water and salt from your body. The high blood pressure that results can then make kidney disease worse. Worsening kidney disease can raise blood pressure again. A dangerous cycle results as each disease makes the other worse. More information is provided in the NIDDK health topic, [High Blood Pressure and Kidney Disease](#).

Who gets kidney disease?

You are more likely to develop kidney disease if you have

- diabetes
- high blood pressure

- heart disease
- lupus
- a family member who has kidney failure

African Americans, Hispanics/Latinos, American Indians, and Alaska Natives are also more likely to develop kidney disease.

Who gets heart disease?

You are more likely to develop heart disease if you have

- high blood pressure
- high blood cholesterol, a blood fat
- diabetes
- kidney disease
- a family history of early heart disease

You are also more likely to develop heart disease if you

- smoke
- eat a diet high in the wrong kinds of fats
- have too much sodium—often from salt—in your diet
- are overweight
- don't exercise
- are a man age 45 or older
- are a woman age 55 or older

What are the tests for kidney disease?

Tests for kidney disease include a blood test and a urine test. Both tests can be done in your doctor's office or at a commercial facility.

- The blood test checks your GFR, which tells how well your kidneys are filtering. GFR stands for glomerular filtration rate.
- The urine test checks for albumin in your urine. Albumin is a protein that can pass into the urine when the kidneys are damaged.

If these tests show you have kidney disease, you may need to repeat these tests on a regular basis. More information is provided in the NIDDK health topic, [Testing for Kidney Disease](#).

What are the tests for heart disease?

You can get several routine tests for heart disease in your doctor's office or at an outpatient center, including

- a blood pressure test
- a blood test for cholesterol
- an electrocardiogram—a test using sensors stuck to your skin to record the electrical signals that keep your heart beating

Another test, called a stress test, measures how your heart works during physical activity, such as walking, running on a treadmill, or riding a bike. This test usually takes place in a hospital or a specialized outpatient center.

If these tests show that you have heart disease, your doctor may want to do more tests.

How are kidney disease and heart disease treated?

If you have kidney disease or heart disease, your health care provider will want you to control your blood glucose if you have diabetes and your blood pressure if you have high blood pressure. Your health care provider may prescribe medicines to keep your blood glucose and blood pressure under control. Medicines for blood glucose control can include insulin injections and pills. Certain medicines for blood pressure may also keep your kidney disease from getting worse.

Another important part of treating kidney disease and heart disease includes living a healthy lifestyle. Staying active can help protect your kidneys and your heart. Try to be active for 30 minutes or more most days of the week. Start with easy activities such as walking slowly or raking leaves. Later, try some activities that get your heart pumping, such as walking briskly or swimming. Always talk with your health care provider before starting any new exercise program. You can find more information about about weight control and physical activity at NIDDK's [Weight Control Information Network](#). See the "[Eating, Diet, and Nutrition](#)" section below for tips on eating a healthy diet that will protect your heart and kidneys.

How can I prevent kidney disease and heart disease?

You cannot always prevent kidney disease and heart disease. However, you can lower your chance of having kidney disease and heart disease by taking the following steps:

- See your health care provider as directed.
- Keep your blood pressure below 140/90. Follow your provider's advice on how to stay at or below your target.
- Control your blood glucose if you have diabetes.

- Have your blood and urine checked as your provider instructs.
- Try to keep your cholesterol numbers in a healthy range. Talk with your provider about your cholesterol goals.
- Control your weight. If you are overweight, talk with your provider about how you can lose weight.
- Be physically active 30 minutes a day most days of the week.
- Take all medicines as prescribed.
- Eat healthy—see “Eating, Diet, and Nutrition.”

Eating, Diet, and Nutrition

A healthy diet can help protect you from heart disease and kidney disease. Keep your heart and kidneys healthy by eating plenty of the following foods:

- fruits and vegetables
- whole-grain breads and cereals
- low-fat milk and milk products such as yogurt and cheese
- lean meats or meat substitutes such as tofu
- fish
- unsaturated fats such as olive oil or corn oil
- low-sodium foods

People with *advanced* chronic kidney disease may need to adjust their diet to avoid high potassium. If you have advanced kidney disease, you may need to limit foods such as bananas, oranges, potatoes, and tomatoes and eat apples, berries, grapes, and peaches instead. Check with your provider to find out if you should cut back on your potassium. Do not alter your diet without checking because you might eat less of these healthy fruits and vegetables unnecessarily.

More information is provided in the NIDDK health topic, [Potassium: Tips for People with Chronic Kidney Disease](#).

You need protein. However, protein breaks down into wastes that your kidneys need to remove. Most people eat more protein than they need. Large amounts of protein make your kidneys work harder. High-quality proteins such as meat, fish, and eggs make fewer wastes than other sources of protein. Beans, whole grains, soy products, nuts and nut butters, and dairy products can also be good sources of protein.

More information is provided in the NIDDK health topic, [Protein: Tips for People with Chronic Kidney Disease](#).

Phosphorus is a mineral that helps keep your bones healthy. Phosphorus also helps blood vessels and muscles work. Phosphorus is found naturally in foods rich in protein, such as meat, poultry, fish, nuts, beans, and dairy products. Phosphorus is also added to many processed foods. When you have kidney disease, phosphorus can build up in your blood, making your bones thin, weak, and more likely to break. Many people with kidney disease need to eat foods with less phosphorus than they are used to eating.

More information is provided in the NIDDK health topic, [Phosphorous: Tips for People with Chronic Kidney Disease](#).

Sodium is a part of salt. Sodium is found in many canned, packaged, and “fast” foods. Sodium is also found in many seasonings and meats. You should limit the amount of sodium you get to 2,300 mg or less each day.

More information is provided in the NIDDK health topic, [Sodium: Tips for People with Chronic Kidney Disease](#).

A dietitian is a health care provider who helps people with kidney disease choose the right foods and plan healthy meals. Talk with a dietitian about foods that might harm you and foods you should add to your diet.

More information is provided in the NIDDK health topic, [Eating Right for Kidney Health: Tips for People with Chronic Kidney Disease](#).

What should I ask my doctor about kidney disease and heart disease?

Some questions to ask your doctor include

- What is my blood pressure? What should it be?
- How often should I have my blood pressure checked?
- Should I be taking medicines to control my blood pressure?
- Should I be on a special diet?
- What are my cholesterol numbers? What should they be?
- How much exercise should I be getting?
- Where can I get help to quit smoking?

Points to Remember

- If you have kidney disease, you are more likely to get heart disease. Heart disease is the most common cause of death among people who have kidney disease.

- Kidney disease means that the kidneys are damaged and can't filter blood as they should. This damage can cause wastes to build up in the body.
- Heart disease includes any problem that keeps your heart from pumping blood as well as it should. The problem might start in your blood vessels or your heart.
- Tests for kidney disease include a blood test and a urine test.
- You can get several routine tests for heart disease in your doctor's office or at an outpatient center, including
 - a blood pressure test
 - a blood test for cholesterol
 - an electrocardiogram
 - a stress test
- If you have kidney disease or heart disease, you will want to control your blood glucose if you have diabetes, and your blood pressure if you have high blood pressure.
- You can lower your chance of having kidney disease and heart disease by taking the following steps:
 - See your health care provider as directed.
 - Keep your blood pressure below 140/90. Follow your provider's advice on how to stay at or below your target.
 - Control your blood glucose if you have diabetes.
 - Have your blood and urine checked as recommended.
 - Try to keep your cholesterol numbers in a healthy range.
 - Control your weight. If you are overweight, talk with your provider about how you can lose weight.
 - Be physically active 30 minutes a day most days of the week.
 - Take all medicines as prescribed.
 - Eat healthy.
- You should limit the amount of sodium you get to 2,300 mg or less each day. Sodium is a part of salt.